Lecture 3: NuMicro MCU Learning Board Introduction

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Outline

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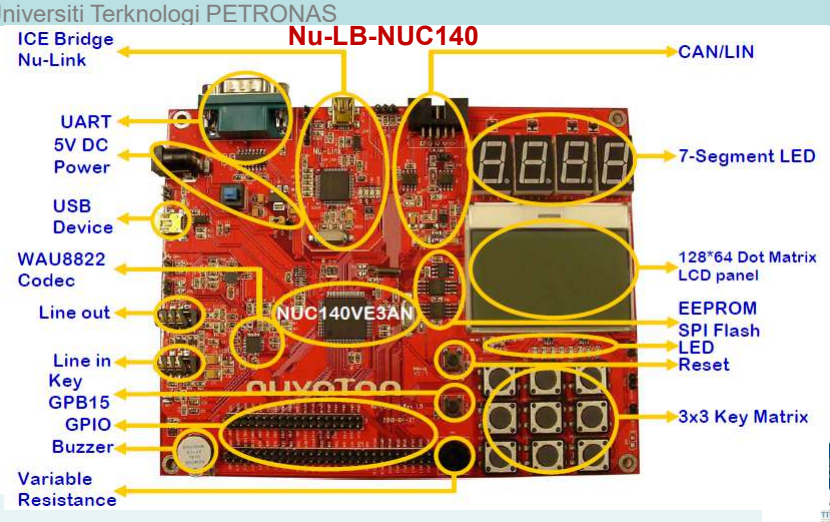
Learning Board Supply Voltage is **+5 Volt**

- Learning Board Connectors
- Learning Board Pin used and its function
- Learning Board Schematics

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NuMicro MCU Learning Board

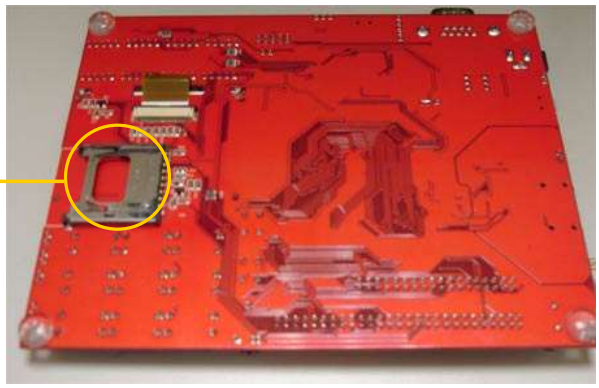
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NuMicro MCU Learning Board (bottom view)

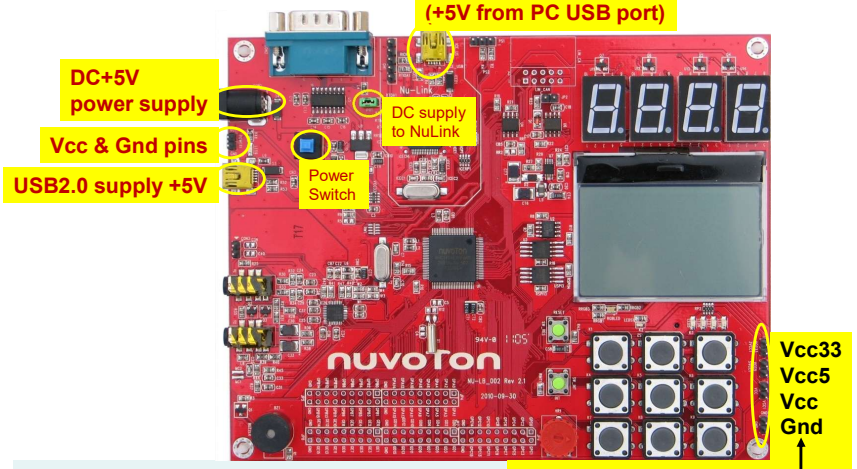
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Power Connectors

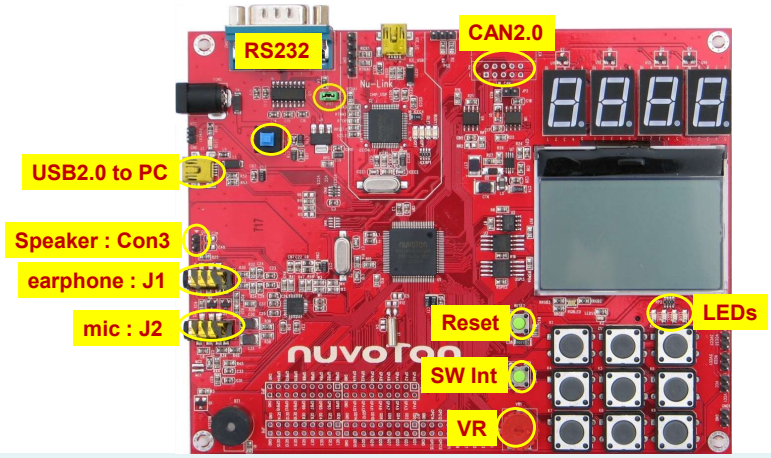
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Peripheral Connectors & Buttons

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Notices of Power Failure

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- PC USB port supply maximum current of **500mA**

- When **shorted circuits** or **over-drive current** from external electronic circuits or peripheral modules, PC may cut off the power of the USB port

→ user will require to **reboot the PC** to regain power of the USB port

- NUC140 chip is very **hot** when the learning board is running

→ The board may be **damaged** due to pin mis-connected to high voltage (>5V)

→ The learning board should be **replaced.**

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Notices of LCD no display

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- LCD has no display when it is cracked or damaged

→ **Replace the LCD**

- LCD has no display when its soft cable on the back of the board was unplugged and did not reconnect properly

→ User need **to reconnect the LCD soft cable** to the interface socket tightly

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Pin named by GPIO groups

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Pin named by GPIO groups A~E

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- **NuMicro MCU Learning Board has 5 group of pins : GPA, GPB, GPC, GPD, GPE**
- **Pins are named by GPA, GPB, GPC, GPD, GPE with a number 0~15**
  - **E_GPA, E_GPB, E_GPC, E_GPD, E_GPE as its defined port name in the C code**
- **To connect to external sensors / circuits, you can jump wire from these pins to the external devices**

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GPIO group A, B

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JP4: GPIOA

GPA	Define		
1	GPIOA0	ADC0	
2	GPIOA1	ADC1	
3	GPIOA2	ADC2	
4	GPIOA3	ADC3	
5	GPIOA4	ADC4	
6	GPIOA5	ADC5	
7	GPIOA6	ADC6	
8	GPIOA7	ADC7	SPI_SS21
9	GPIOA8	I2C0SDA	
10	GPIOA9	I2C0SCL	
11	GPIOA10	I2C1SDA	
12	GPIOA11	I2C1SCL	
13	GPIOA12	PWM0	
14	GPIOA13	PWM1	
15	GPIOA14	PWM2	
16	GPIOA15	PWM3	
17-18	GND		

JP5 : GPIOB

GPAB	DEFINE		
1	GPIOB0	RX0	
2	GPIOB1	TX0	
3	GPIOB2	RTS0	
4	GPIOB3	CTS0	
5	GPIOB4	RX1	
6	GPIOB5	Tx1	
7	GPIOB6	RTS1	
8	GPIOB7	CTS1	
9	GPIOB8	TM0	STADC
10	GPIOB9	TM1	SPI_SS11
11	GPIOB10	TM2	SPI_SS01
12	GPIOB11	TM3	
13	GPIOB12	CPO0	
14	GPIOB13	CPO1	
15	GPIOB14	INT0	SPI_SS31
16	GPIOB15	INT1	
17-18	GND		

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GPIO group C, D

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JP6 : GPIOC

GPCD	DEFINE	
1	GPIOC0	SPI_SS00
2	GPIOC1	SPICLK0
3	GPIOC2	SDI00
4	GPIOC3	SDO00
5	GPIOC4	SDI01
6	GPIOC5	SDO01
7	GPIOC6	CPP0
8	GPIOC7	CPN0
9	GPIOC8	SPI_SS10
10	GPIOC9	SPICLK1
11	GPIOC10	SDI10
12	GPIOC11	SDO10
13	GPIOC12	SDI11
14	GPIOC13	SDO11
15	GPIOC14	CPP1
16	GPIOC15	CPN1
17-18	GND	

JP7 : GPIOD

GPCD	DEFINE	
1	GPIOD0	SPI_SS20
2	GPIOD1	SPICLK2
3	GPIOD2	SDI20
4	GPIOD3	SDO20
5	GPIOD4	SDI21
6	GPIOD5	SDO21
7	GPIOD6	CAN0_RX
8	GPIOD7	CAN0_TX
9	GPIOD8	SPI_SS30
10	GPIOD9	SPICLK3
11	GPIOD10	SDI30
12	GPIOD11	SDO30
13	GPIOD12	SDI31
14	GPIOD13	SDO31
15	GPIOD14	CAN1_RX
16	GPIOD15	CAN1_TX
17-18	GND	

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GPIO group E

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JP8 : GPIOE

GPE	DEFINE
1	GPIOE0
2	GPIOE1
3	GPIOE2
4	GPIOE3
5	GPIOE4
6	GPIOE5
7	GPIOE6
8	GPIOE7
9	GPIOE8
10	GPIOE9
11	GPIOE10
12	GPIOE11
13	GPIOE12
14	GPIOE13
15	GPIOE14
16	GPIOE15
17-18	GND



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Nu-LB-NUC140 users manual

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- The user's manual includes
  (filename: 1.Nu-LB-NUC140_User_s_Guide_v2.0.pdf)
  - Board Configurations
  - Board Functions and the related pins of MCU
  - Board Schematics



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Nu-LB-NUC140 pin used & its function

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Block	Pin	Function
ICE Bridge Nu-Link	ICE_CLK ICE_DATA	SWD interface
UART	GPB0 GPB1	UART0 Rx UART0 Tx
WAU8822 codec	GPC0 GPC1 GPC2 GPC3 GPA15	I2SLRCLK I2SBCLK I2SDI I2SDO I2SMCLK
	GPA8 GPA9	I2C0 SDA I2C0 SCL
	GPE14	Line out Enable/Disable
	GPE15	Line in plug in/out detect
EXT INT	GPB15	Software Interrupt

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Nu-LB-NUC140 pin used & its function

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Block	Pin	Function
CAN	GPD6 GPD7	CAN0 Rx CAN0 Tx
	GPB12~13	CAN transceiver speed
LIN	GPB4 GPB5	UART1 Rx UART1 Tx
	GPB6	LIN transceiver wakeup function
	GPB7	LIN transceiver Enable/Disable
7-Seg LED	GPE0~7	Row
	GPC4~7	Column
Black Dot Matrix LCD Panel	GPD8 GPD9 GPD10 GPD11`	SPI3 SS30 SPI3 SPCLK LCD Reset SPI3 MOSI0
	GPD14	Backlight Enable/Disable

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Nu-LB-NUC140 pin used & its function

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Block	Pin	Function
Variable Resistor	GPA7	ADC interface
Buzzer	GPB11	PWM4
Keypad	GPA0~5	GPIO
Reset Button	RESET	Reset
EEPROM	GPA10 GPA11	I2C1 SDA I2C1 SCL
FLASH	GPD0 GPD1 GPD2 GPD3 GPD4 GPD5	SPI2 SS20 SPI2 SPCLK SPI2 MISO0 SPI2 MOSI0 SPI2 MISO1 SPI2 MOSI1

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Nu-LB-NUC140 pin used & its function

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Block	Pin	Function
LED	GPA12 GPA13 GPA14	PWM0 PWM1 PWM2
	GPC12~15	GPIO
SD card	GPD12 GPD13	SD card Power Enable/Disable SD card insert detection
	GPC8 GPC9 GPC10 GPC11	SPI1 SS10 SPI1 SPCLK SPI1 MOSI0 SPI1 MISO0

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Learning Board Schematics

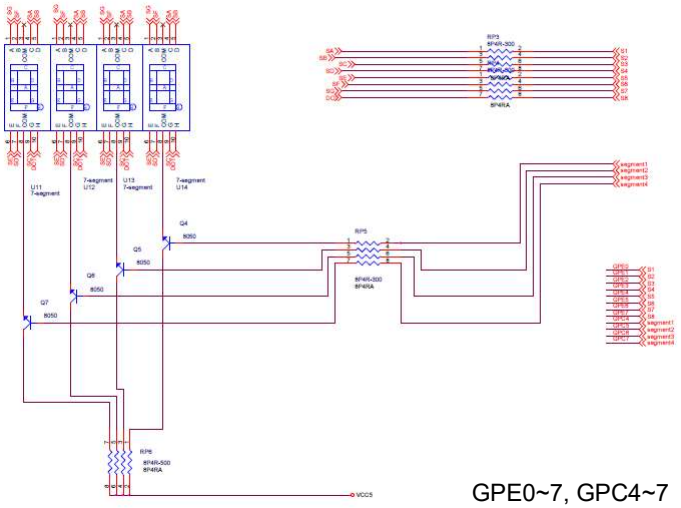
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- 7-segment schematic
- 64x128 LCD schematic
- RGB LED schematic
- Keypad & SW_INT schematic
- UART/RS232 schematic
- I2C & Buzzer schematic
- SPI schematic
- PS2 & ADC schematic
- I2S schematic
- CAN&LIN schematic
- SD card schematic

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7 segment schematic

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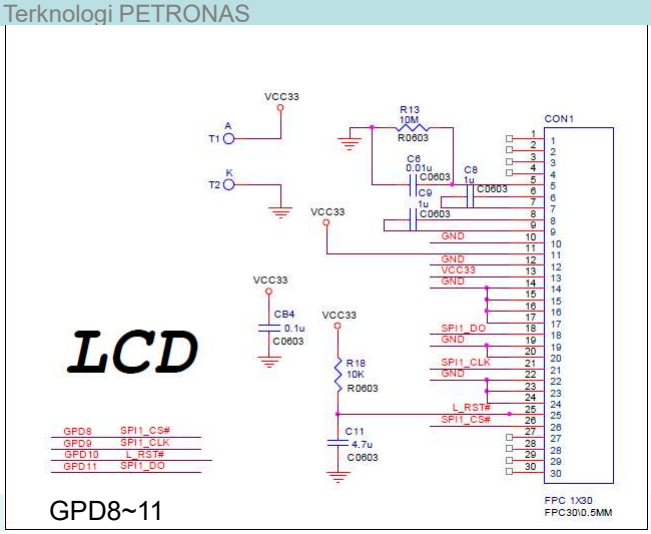


GPE0~7, GPC4~7

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64x128 LCD schematic

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GPD8~11

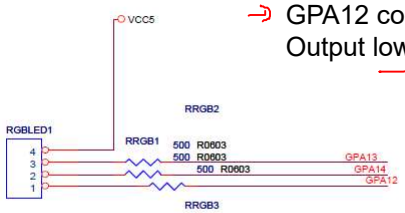
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RGB LED schematic

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- GPA14 control Blue color
- GPA13 control Green color
- GPA12 control Red color

  Output low to enable R/G/B LEDs

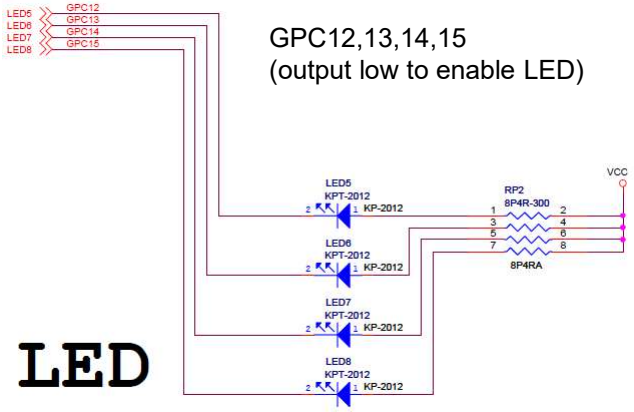


RGB LED

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Red LED schematic

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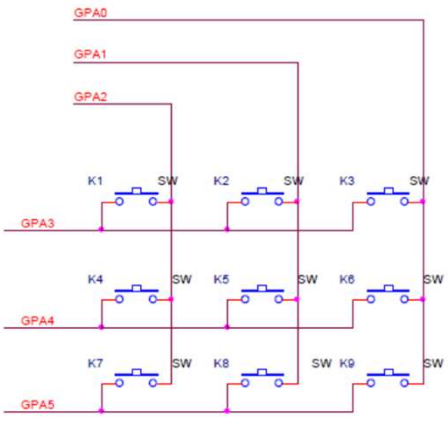
GPC12,13,14,15
(output low to enable LED)

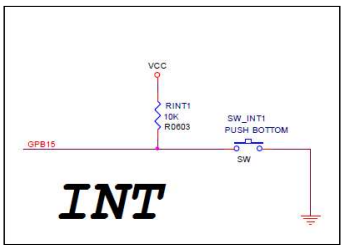
LED

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Keypad & SW_INT schematic

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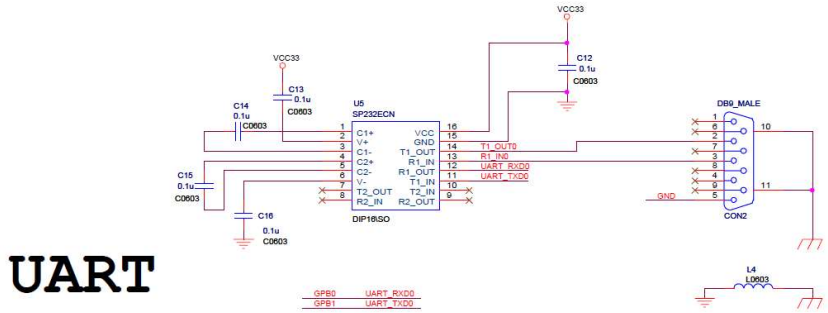


GPA0~5 GPB15

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UART/RS232 schematic

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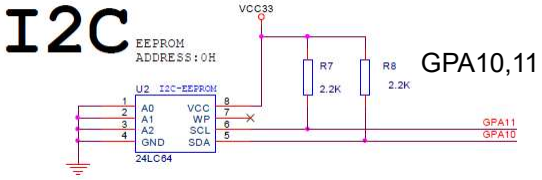
UART

GPB0,1

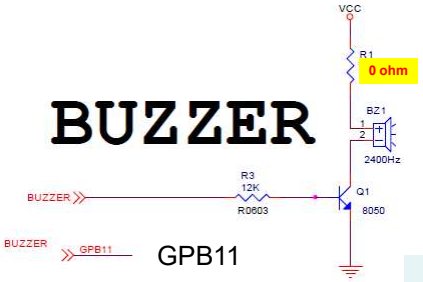
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I2C & Buzzer schematic

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GPA10,11

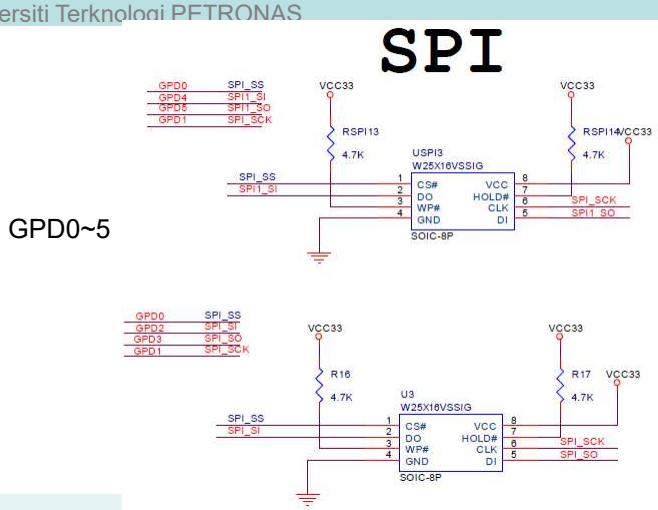


GPB11

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SPI schematic

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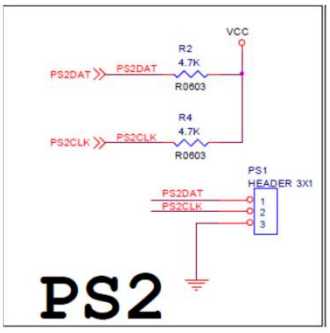


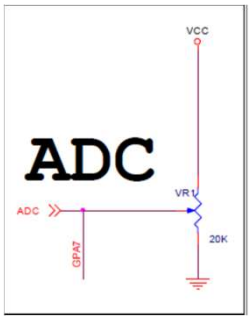
GPD0~5

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PS2 & ADC schematic

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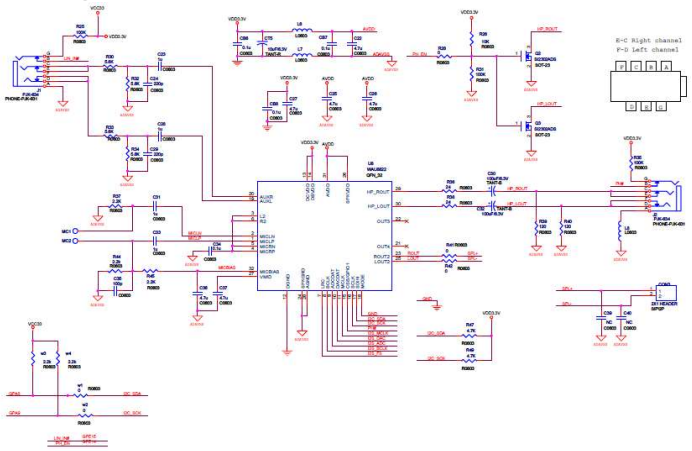


GPA7

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I2S schematic

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GPA8,9

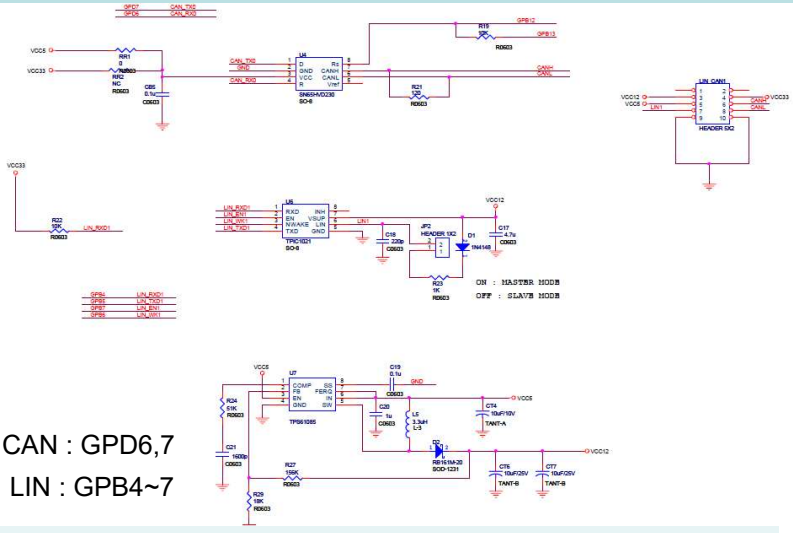
GPA15

GPC0~3

I2S

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CAN&LIN schematic



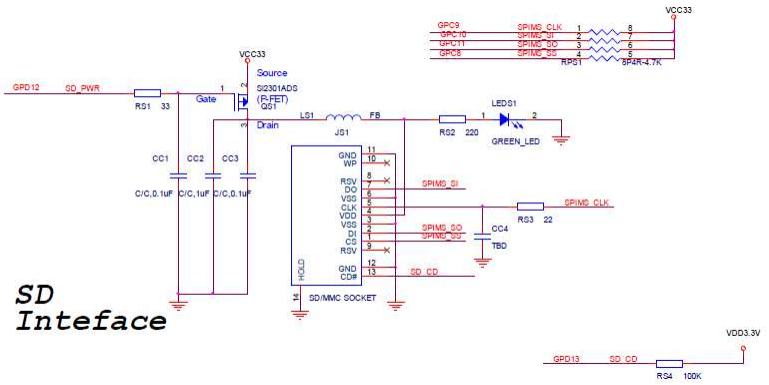
CAN : GPD6,7

LIN : GPB4~7

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SD card schematic

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GPD12,13
GPC8~11



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